

The Hodge Conjecture, Computationally: What Can Be Checked, What Cannot, and What Machine Learning Adds

Topological consistency of 7,890 CICY threefolds, Shioda's Hodge characters of Fermat fourfolds, and learning h^{11} with PINNeAPLe

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Abstract

The Hodge conjecture asserts that on a smooth complex projective variety every rational (p,p) cohomology class is a rational combination of classes of algebraic cycles. It is a statement about the existence of cycles, not about numbers, so it is not falsifiable by the kind of numerical evidence available for the other Millennium problems. We therefore separate three tasks. (i) Consistency: for all 7,890 complete-intersection Calabi–Yau threefolds of the Candelas et al. list, the Euler characteristic computed independently from the configuration matrix (Chern classes and intersection numbers on products of projective spaces) equals $2(h^{11} - h^{21})$ in 7,890/7,890 cases. For threefolds the Hodge conjecture is already a theorem, so this validates data and code, not the conjecture. (ii) The conjecture proper, in a family where it is combinatorially accessible: using Shioda's character description we count the Hodge classes of the Fermat fourfolds of degree $3 \leq m \leq 20$ and how many of them are spanned by products of linear-subspace cycles (pair-decomposable characters). For prime m and $m = 4$ all Hodge characters are pair-decomposable; for composite m a large fraction is not, which is exactly where other constructions are needed (the conjecture has nonetheless been proved for many such degrees, most recently for all odd $m \leq 199$). (iii) Learning: a validation-selected PINNeAPLe MLP predicts h^{11} exactly for 83.1% of held-out CICYs versus 62.6% for the favourable-embedding guess.

Keywords: Hodge conjecture; Calabi–Yau; CICY; Fermat varieties; algebraic cycles; machine learning; Millennium Prize problems

What this study shows — and what it does not

- Shows: exact topological consistency of the CICY Hodge-number list (7,890/7,890) with an independent Chern-class computation.
- Shows: exact counts of Hodge classes of Fermat fourfolds ($3 \leq m \leq 20$) and of the subset reached by the classical linear-cycle argument.
- Does not show anything for or against the Hodge conjecture: no computation here can produce or rule out an algebraic cycle. Non-pair-decomposable characters are NOT counterexamples.

1 Strategy

For each Millennium problem our protocol is the same: identify the strongest statement a computation can actually certify, compute it exactly or with quantified error, and say what remains. For the Hodge conjecture that statement is narrow. We chose (a) a large, published family of varieties to validate our algebraic-geometry pipeline against known Hodge numbers, and (b) the Fermat family, where Hodge classes have an exact combinatorial description (Shioda [3]) and the conjecture has a concrete combinatorial shadow.

2 Methods

2.1 Euler characteristics of CICY threefolds

A CICY $X \subset \mathbb{P}^{n_1} \times \dots \times \mathbb{P}^{n_k}$ is the common zero locus of K polynomials with multidegrees given by a $k \times K$ matrix q . Its total Chern class is $c(X) = \prod_i (1 + J_i)^{n_i+1} / \prod_j (1 + \sum_i q_{ij} J_i)$, and $\chi(X) = \int_X c_3$ is the coefficient of $\prod_i J_i^{n_i}$ in $c_3 \cdot \prod_j (\sum_i q_{ij} J_i)$. We expand $c(X)$ to degree 3 exactly (integer arithmetic) and extract the top coefficient by memoised recursion over the linear factors. For a Calabi–Yau threefold $\chi = 2(h^{11} - h^{21})$.

2.2 Hodge characters of Fermat fourfolds

For $X_m: x_0^m + \dots + x_5^m = 0$, the primitive middle cohomology decomposes into one-dimensional eigenspaces indexed by $\alpha = (a_0, \dots, a_5)$, $a_i \in \mathbb{Z}/m \setminus \{0\}$, $\sum a_i = 0$; α lies in $H^{p,q}$ with $p = |\alpha| - 1$, $|\alpha| = \sum \langle a_i \rangle / m$. The rational Hodge classes are spanned by the Galois orbits of characters with $|\alpha| = 3$ for every unit $t \in (\mathbb{Z}/m)^\times$ [3]. A character is pair-decomposable if its entries split into pairs with $a_i + a_j \equiv 0$; these are the classes obtained from linear subspaces. Built-in checks: the method reproduces 19 primitive Hodge classes for the Fermat quartic surface (Picard number 20) and 6 for the cubic surface (Picard number 7).

2.3 Learning h^{11}

Input: the configuration matrix zero-padded to 12×15 plus the number of projective factors; output: h^{11} as a class. 70/15/15 random split. Candidates from the PINNeAPLe model registry (GenericMLP, GenericResMLP, ModifiedMLP) are trained with early stopping; the one with the best validation accuracy is the only one scored on test. A first attempt with a single Fourier-feature ModifiedMLP (bandwidth fixed a priori) is reported as well: it failed.

3 Results

CICY consistency. 7,890 of 7,890 configurations satisfy $\chi = 2(h^{11} - h^{21})$ exactly; there are 0 disagreements.

Table 1. Hodge characters (dimension over $\mathbb{C}$ of the primitive rational Hodge classes, counted character by character) of Fermat fourfolds.

m	Hodge characters	pair-decomposable	not pair-decomposable	fraction reached
3 (prime)	20	20	0	1.000
4	141	141	0	1.000
5 (prime)	400	400	0	1.000
6	1,751	1,001	750	0.572
7 (prime)	1,860	1,860	0	1.000
8	5,881	3,301	2,580	0.561
9	8,000	5,120	2,880	0.640
10	19,881	7,761	12,120	0.390
11 (prime)	10,900	10,900	0	1.000
12	52,991	15,101	37,890	0.285
13 (prime)	19,920	19,920	0	1.000
14	77,401	26,041	51,360	0.336
15	78,260	32,900	45,360	0.420
16	87,921	41,301	46,620	0.470
17 (prime)	50,560	50,560	0	1.000
18	264,671	61,601	203,070	0.233
19 (prime)	73,620	73,620	0	1.000
20	346,381	87,661	258,720	0.253

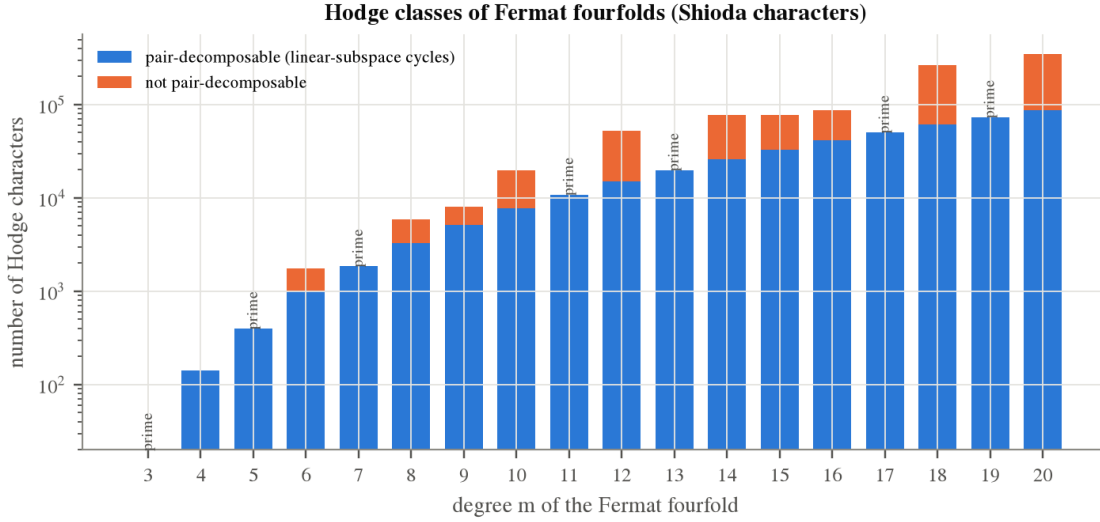


Figure 1. Hodge characters of Fermat fourfolds, split into those reached by products of linear cycles and the rest (log scale).

For every prime degree the linear-cycle classes exhaust the Hodge classes, recovering Shioda’s theorem for prime m ; for $m = 4$ as well. For composite m the fraction drops quickly (25% at $m = 20$). Those classes require other cycles (e.g. the Aoki–Shioda constructions); a 2026 computer-assisted proof covers every odd $m \leq 199$ [6]. Our counts are an exact, reproducible census, not a test.

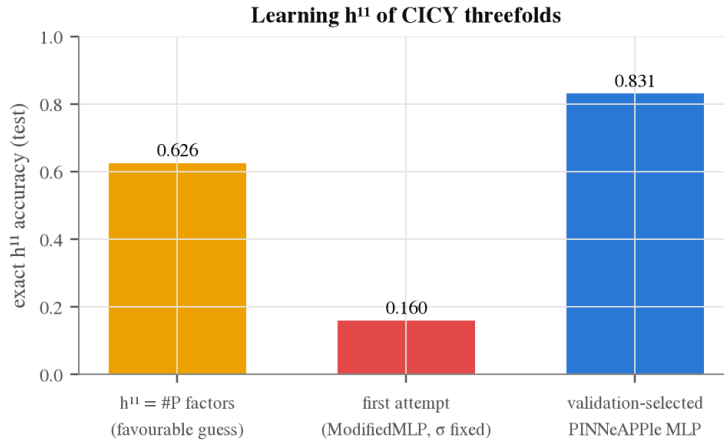


Figure 2. Exact- h^{11} accuracy on held-out CICYs.

Table 2. Model selection on validation. Selected: bench_mlp(w256,d4,relu); test accuracy 0.831; on the non-favourable test CICYs ($h^{11} \neq \text{number of factors}$) 0.666.

Candidate (PINNeAPLe registry)	Validation accuracy
bench_mlp(w128,d3,relu)	0.787
bench_mlp(w256,d4,relu)	0.822
bench_res_mlp(w128,d4)	0.817
modified_mlp(sigma0.1)	0.175

4 Discussion

The honest summary is that computation can certify consistency and count Hodge classes exactly, and learning can reproduce known invariants; none of these touches the existence of algebraic cycles, which is the content of the conjecture. The one lesson for the PINNeAPLe toolchain is methodological: Fourier-feature networks designed for smooth fields generalise poorly on discrete configuration data; plain MLPs selected on validation are the right default there.

References

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